Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21325

F-425

B.C.A. (Fifth Semester) Examination, 2025-26

(New Course)

Paper - BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Explain the need for parallelism in modern computer systems.

   (b) Discuss different models of parallel computation and their relevance in current architectures.

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2. (a) Compare cache write strategies: Write-through, Write-back, and Write-around.

   (b) Explain the different memory addressing schemes used in parallel computers.

3. (a) Illustrate the stages of an Instruction Pipeline with a neat diagram.

   (b) What is pipeline interlocking? Explain methods used to handle interlocks.

4. (a) Explain the architecture and working of a Vector Processing system with suitable examples.

   (b) Discuss Vector Chaining and its impact on system performance.

5. (a) Write short notes on the following (any two):

   1. Shuffle Exchange Network
   2. SIMT Architecture
   3. Barrel Shifter

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(b) Explain the algorithms used for matrix addition in SIMD processors.

6. (a) Differentiate between centralized and distributed shared memory models.

(b) Explain the structure and role of Inter-connection Networks in multiprocessors.

7. (a) Describe the UMA, NUMA, and NORMA memory models with their characteristics

(b) Explain tightly coupled multiprocessors and discuss their advantages.

8. Write in brief about the following (any four):

(a) Out-of-order execution

(b) VLIW architecture

(c) Snoopy cache protocol

(d) Flynn's taxonomy

(e) Multiprocessor scheduling

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